

Embedded System & Internet of Things(IoT)

Course By

The IoT Academy

Duration- 6 Months

About Us

We are a fast emerging company imparting quality and affordable programs for colleges and individuals for digital economy skills training, internship and guidance focused on helping people develop the skills they need to thrive in the rapidly growing digital economy. **The IoT Academy** has made a niche name for itself providing rigorous online/offline training in **Data Science, Machine Learning, AI, Python, Java, Internet of Things (IoT), Embedded Systems, Big Data, Data analytics, Industrial IoT, Industry 4.0, Digital marketing, Etc.** Based in Delhi NCR since 2017 we have helped 150+ professionals, 600+ students and 100+ faculties across many states in India.

We have an easy and affordable learning solution that is accessible to millions of learners. With our learners spread across countries like the US, India, UK, Canada, Singapore, Australia, Middle East, Brazil, and many others, we have built a community of over 1 million learners across the globe.



Our Mission

To upskill 5,00,000 young learners to have their career excellence and create Entrepreneurs in next 5 years.



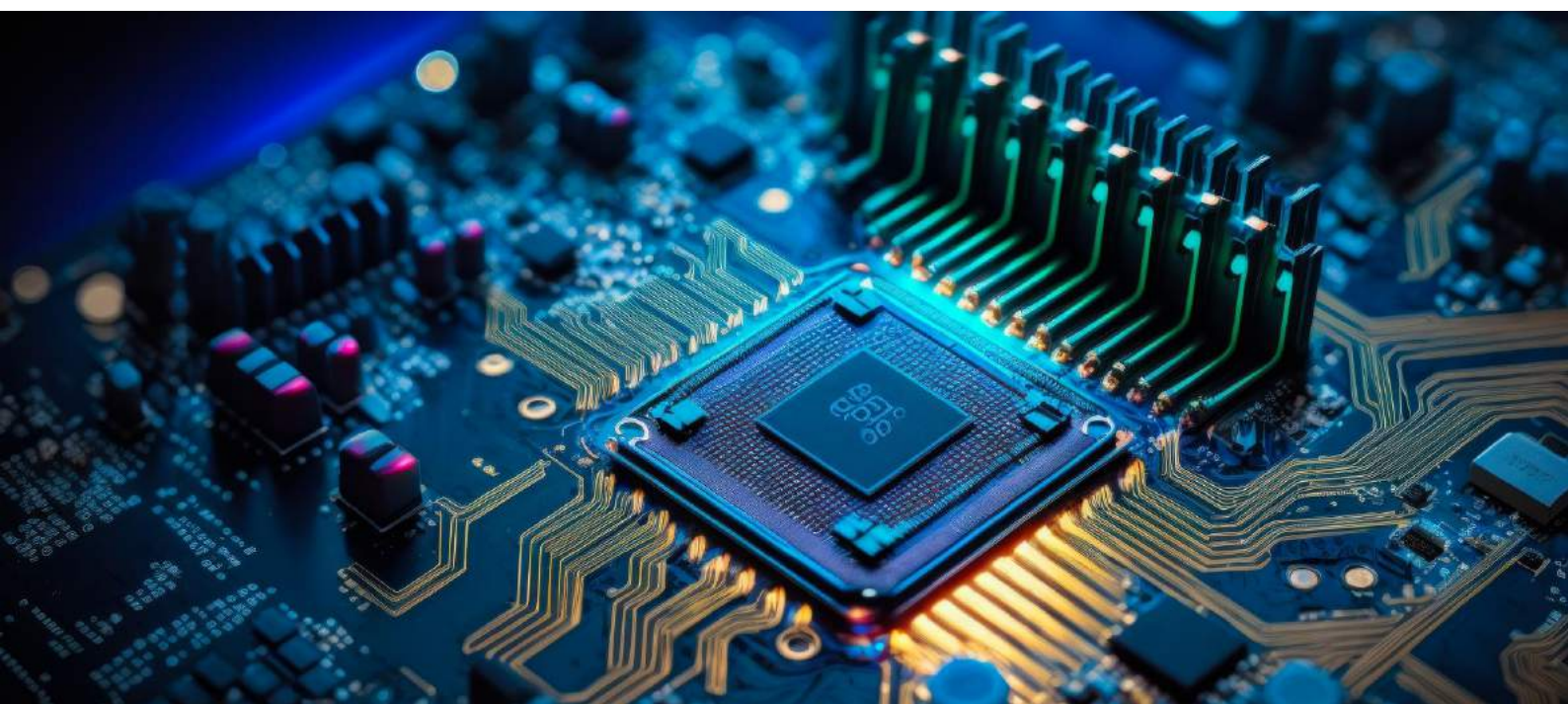
Our Vision

To Impart and contribute significantly in journey of India's 10 Trillion Dollar economy by 2030 by creating high quality Innovations, ambitious and rightly focussed resources.



ABOUT THE PROGRAM

This comprehensive course on Embedded Systems and Internet of Things (IoT) provides students with a deep understanding of the principles, design methodologies, and practical applications of embedded systems and IoT technologies. The course delves into the exciting realm of IoT, exploring the architecture and components that enable devices to connect and communicate over the internet. You will study wireless communication protocols, cloud computing, data analytics, and security mechanisms relevant to IoT applications. Practical projects will guide students in developing IoT-enabled devices, implementing data collection and analysis, and ensuring robust security measures.



Recap of Embedded C

- Datatypes
- Array
- Conditional Statements
- Functions / Callback function
- Structures
- Pointers
- Storage classes

Python

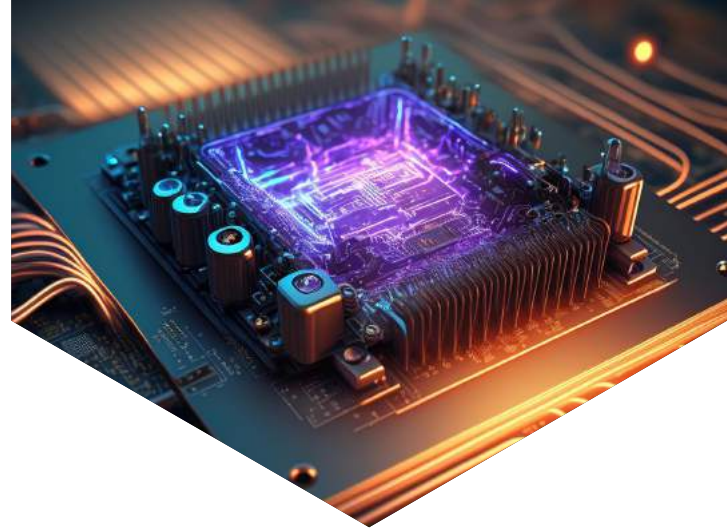
- Data types and type conversion
- Variables and basic operations
- Flow control
- Loops
- Lists, Set and Tuple
- Dictionary
- Functions
- File Handling
- Class and Objects
- Error/ Exception Handling

RPI

- Introduction to Raspberry pi
- Setup and installation
- Basic Led blink
- PWM motor with R-PI
- Switch with R-PI
- Raspberry pi OS walkthrough
- Code execution and data generation

Signal And Acquisition

- Analog and Digital
- Industrial Terminology



Embedded System

- Basic Concepts
- Sensors/Actuators
- Microcontroller units and Architecture (Arduino)
- Interfaces (serial port, SPI, I2C, UART)
- ADC, Timers, RTC, Interrupts, Polling, DMA
- Memory architecture and handling (stack, heap, cache)
- Application driven Selection of Microcontrollers

Communication

- Introduction to communication architecture - Network protocol stack
- Different protocols
- RF: ZigBee, BlueTooth, BLE, Zwave.
- Communication Channels: GSM/ GPRS, WiFi, LoRa & LoRaWAN
- NB-IoT, Comparison between different RF Technologies.
- IPv4 addressing problem for IOT and introduction to IPV6

Arm Microcontroller (STM32)

- Introduction to Microcontrollers (Based on Architecture Selected)
- ARM microcontroller Architecture
- Instruction set architecture - pipelines
- Registers banks/mode and states-theory
- Clock and reset systems, timers and Memory Organisation
- IDE Configuration, Linker Script, Compiler Optimization Options
- Exception/Interrupt handling
- C and Embedded C Migration
- Memory /IO and peripherals - (I2C/SPI /UART/GPIO/Timer/PWM/ADC)
- Debugging tools like SWV debugger

Embedded Linux

LINUX INTERNALS

- Linux internals
- Linux introduction and installation.
- Linux Shell Commands
- Shell Scripting
- C Programming in Linux
- Make Files
- Process Management
- File Operation
- Signals in Linux
- Linux Scheduler & Memory Management
- Linux Multi-Threading Programming
- Inter Process Communication (Pipes, Semaphores, Queues, Shared Memory)
- Network Programming in Linux



SOCKETS

- Socket connection
- Socket Attributes
- Creating a Socket
- Socket Addresses
- Naming a Socket
- Creating a Socket Queue
- Accepting Connections
- Requesting Connections
- Closing a Socket
- Socket Communications Processes and Signals What is process?
- Process Structure
- The Process Table
- Viewing Processes
- System Processes

Inter-Process Communication

- Pipes
- Semaphores, message queues and shared memory
- Shared Memory
- Message Queues

Module-II: Linux Porting

- Introduction, Setup & Hardware
 - Toolchain & Hardware Practical's
 - Bootloader U-Boot
 - U-Boot Porting
 - Customizing Bootloader
 - Linux Kernel
 - Kernel Porting & Compilation
 - Kernel Modification
 - Root File System
 - Embedded Application Development
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Module-III: Linux - Device Drivers

- Introduction and Arch of Linux Device Drivers
 - Kernel Module Programming
 - Character Device Drivers
 - Interrupts in Device Driver
 - Interrupt Handling & Bottom Half
 - Process creation calls (fork,vfork,execv)
 - LDDM (Linux Device Driver Model)
 - Writing and testing of Board File
 - Device Tree
 - Advance Device Drivers
 - Debugging Techniques using GDB & valgrind other tools.
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Module-III: Yocto

- Yocto Architecture
- Recipes defines everything in Yocto
- Layers makes Yocto Modular & Structured
- Adding new Hardware support using BSP Layers
- Custom Distribution & Images
- Creating SDK using Yocto for Application Development

Introduction To IoT

- A What is IoT
- IoT application in different domains
- Trends in IoT Market
- Smart things
- Gate ways
- Middleware
- Edge vs Cloud functional partitioning

IoT Architecture

- Architecture.
- Tech Stack.
- Hardware Development Platforms
- Software Development Platforms
- Communication Protocols
- Power Requirements in IoT
- Cloud, its components and IoT
- Data Streaming and IoT
- Data Store and IoT
- Analytics & Visualization and IoT IoT
- Security

Live Use Cases

- H/W Components
- Client, Device, Rule, Alarm, Device Data, Event
- Sensor data being sent to Cloud platform
- Actuation through BLE
- Visualization through real time dashboard
- How the problem gets solved through big data and visualization





NODEMCU (ESP8266)

- Install IDE Software
- Introduction to NODEMCU (Esp8266)
- Pinout of NODEMCU
- Programming Of NODEMCU
- Weather monitoring solution
- Automatic street lights



NODE-RED

- Installation of Nodejs,
- Installation of Node-Red,
- Building your first flows
- Basic nodes and flows
- A tour of the core nodes
- The Node-RED programming model
- Dashboards and UI techniques
- Local broker installation,
- Connection between local broker and Node-RED

Cloud Computing

- Cloud Computing
 - Benefits of Cloud
 - Deployment Models
 - AWS-IOT Core/ SageMaker
 - Greengrass
 - Google IOT Cloud
 - Thingspeak
 - IBM Bluemix
 - AWS
 - AZURE
 - Google Firebase
 - SQL and NoSQL Data bases
 - Integration with MySQL
 - IoT Cloud Platform: Device Management
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IoT Security

- Introduction to mbed TLS and SSL
 - Importance of IoT application
 - AES encryption basis
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IoT Cloud Services

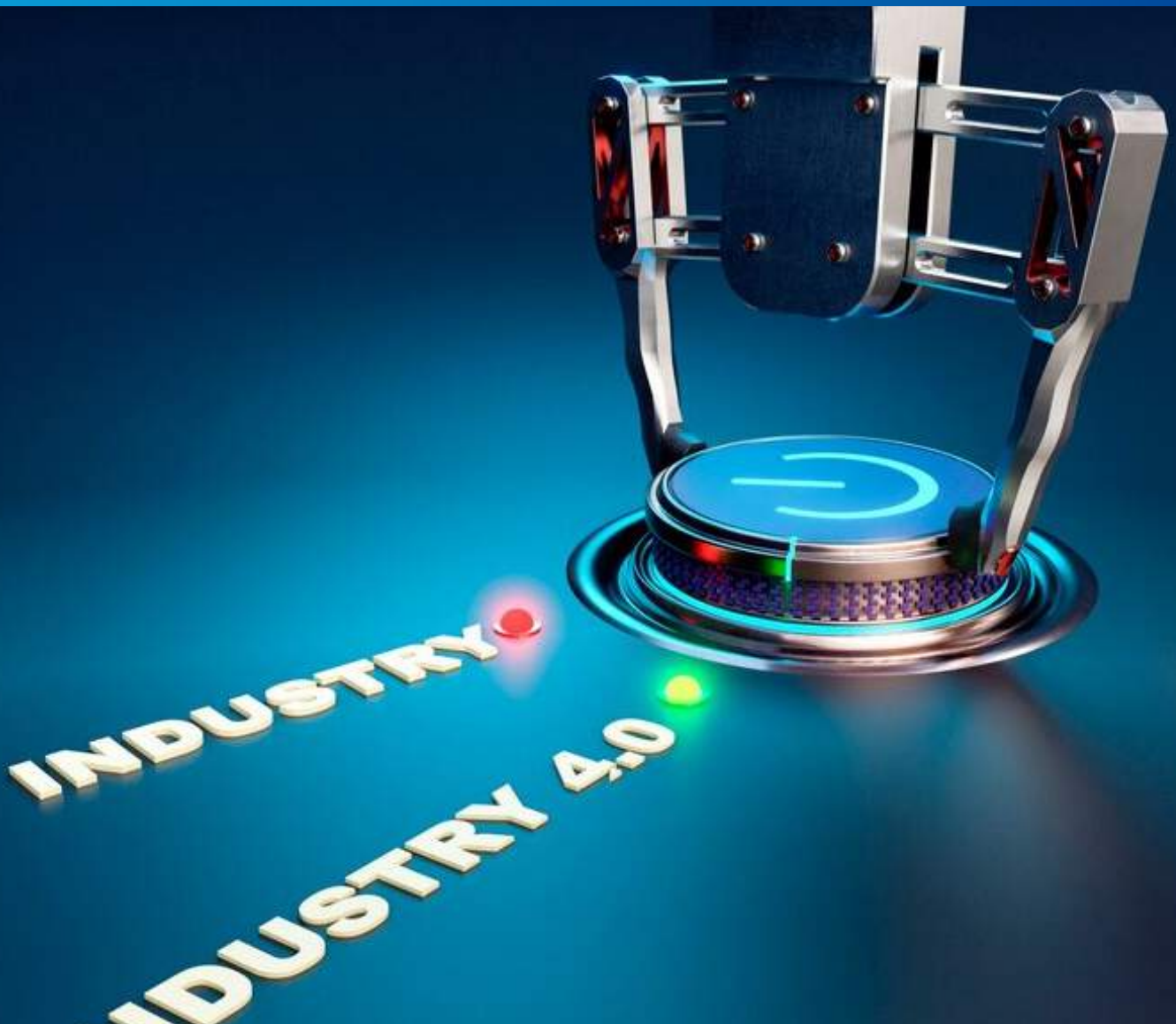
- IoT Cloud Architecture
- Services-SAAS-PAAS-IAAS

Industry 4.0

- Introduction to Industry 4.0
 - Road to Industry 4.0
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Use Cases

- Automobile- basic overview
- Electrical Vehicle



Certification:

Upon successfully completing this program, you'll Learn **Internet of Things (IoT)**.

This certificate will testify to your skills as an expert in **Internet of Things**

Certification ID: xxxxxx



CERTIFICATE OF COMPLETION

This Is To Certify That

Mr./Ms. Name Surname

Is Given The Certificate Of Achievement For Completing The

Internet of Things (IoT)

This Certificate Of Completion Is Being Given In Appreciation Of His/Her Commitment And Efforts And Accumulation Of Knowledge & Skills Which Enabled The Course To Be Completed. We Wish Him/Her Success Throughout His/Her Career.

xxxx 2023 - xxxx 2023

Date



Kaushlendra Singh Sisodia
Chief Mentor, The IoT Academy
Director, Uniconverge Technologies

Thank You!!



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Connecting The Unconnected



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